ProofGuard: an extractor that proves its answers

Project 1: Teach a machine to say "I don't know"

One confident wrong value costs 100 reviews.

So the useful system is not the one that answers most; it is the one that knows when it is right.

1	100	99.01%	41%
human review of a field	a wrong value that gets booked	break-even precision (100/101)	of fields have a value at all: the savings ceiling

Our answer: never trust a value because the model said so. Make every value prove itself, then decide under the real 100:1 costs.

Six layers: prove, challenge, then decide

Layers 1-2 LLM extraction

value + verbatim evidence quote + location | retrieval-first for long contracts, page images for scans

Layer 3 Proof Engine

evidence matching | schema validation | semantic consistency | cross-field + span-boundary checks

Layer 4 H2O-Lightning-4B + adversarial verification

one-token verdict with logprobs: does the evidence support the value? | challenger tries to refute (gray zone only)

Layer 5 Cost-aware risk controller

calibrated P(correct) + 100:1 penalty + bootstrap uncertainty margin (threshold = 20th-percentile saving)

Layer 6 Adaptive router

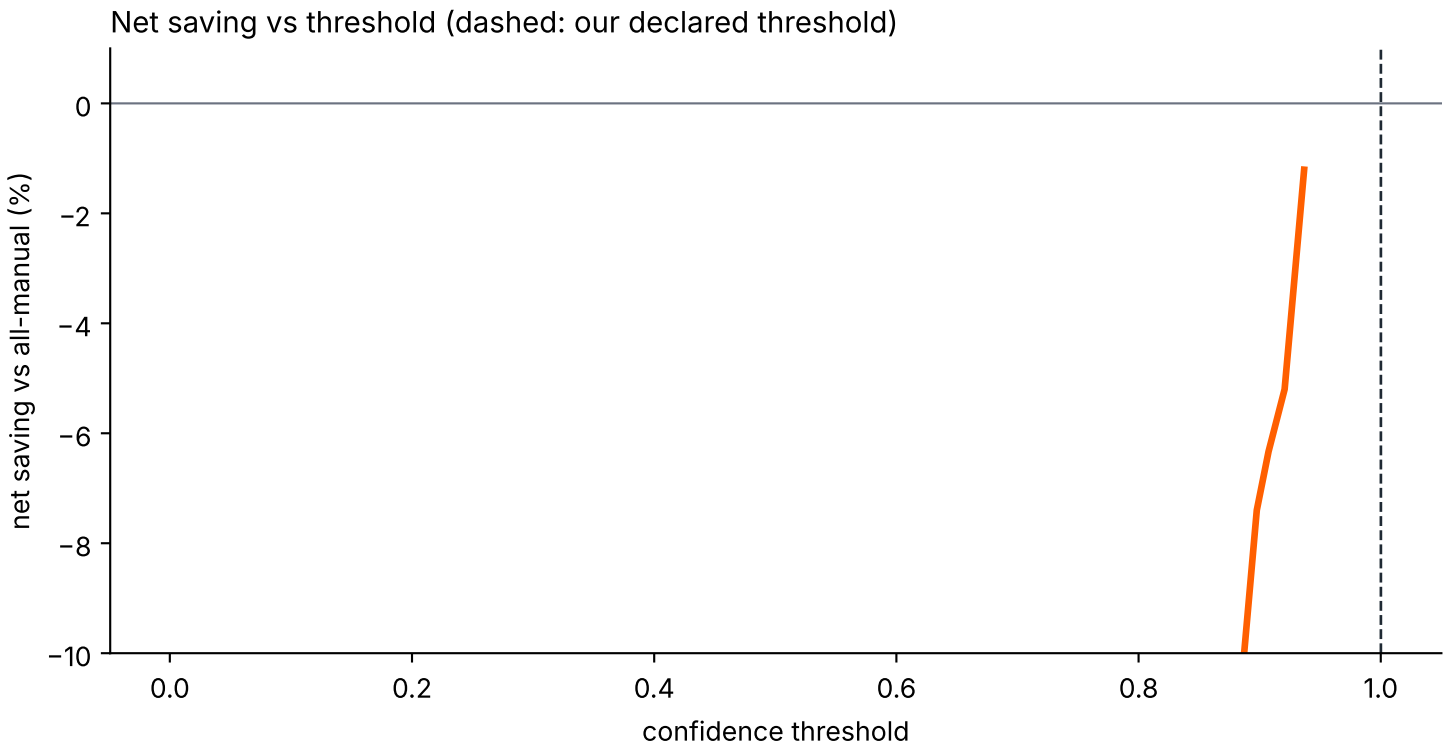
ACCEPT clear cases (cheap) | RECHECK the gray zone (2nd reader + challenger) | REVIEW the rest (a human, cost 1)

Money is spent only where it can change the decision: a clear case costs 1 extraction + 1 one-token check.

Results on documents we never tuned on

Held-out split: 176 documents, 6962 fields. Declared confidence threshold: 1.0001

0.0%	n/a	0.0%	0	0.0%
Net saving	Precision of booked	Coverage	Hallucinations booked	Fields with no human

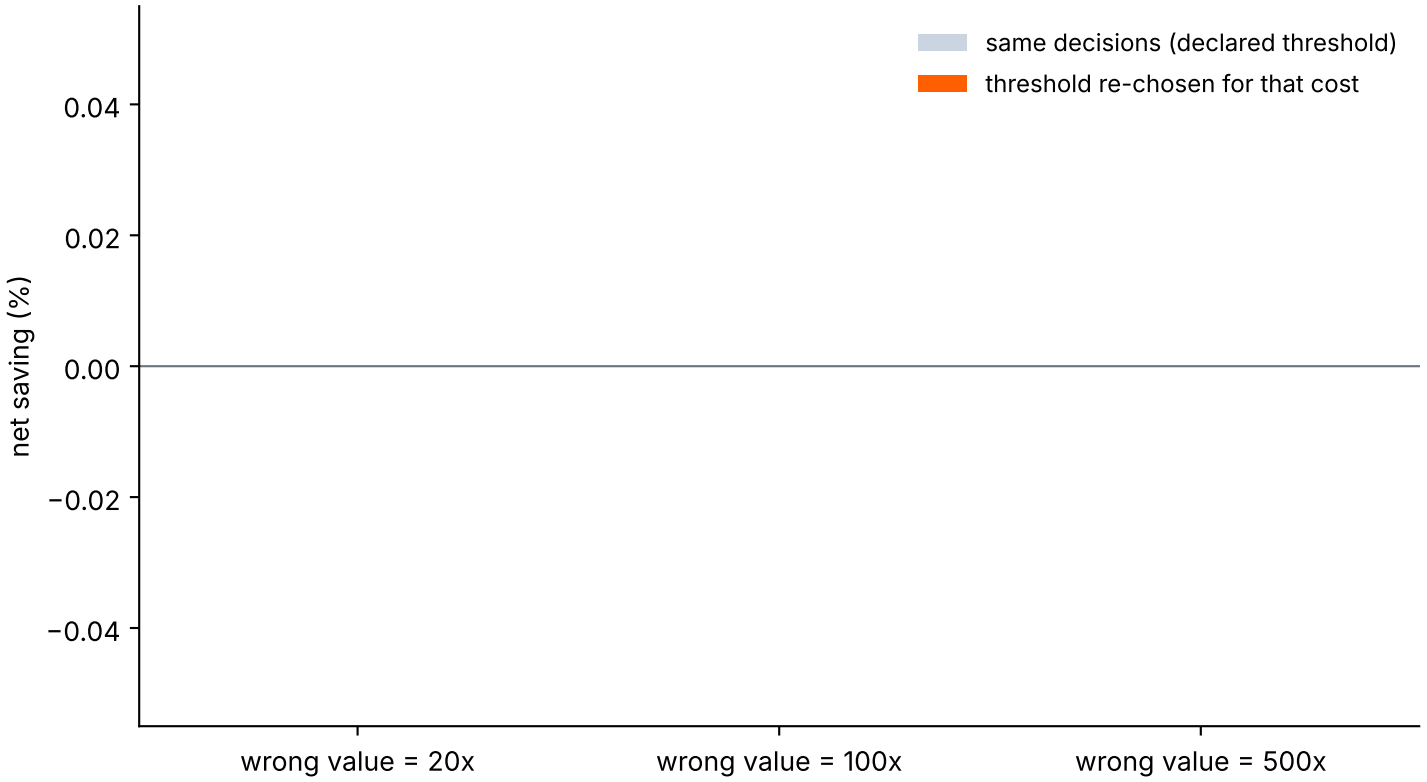


Versus the baselines

ProofGuard (declared threshold)	**0.0%**
Most-common-value baseline (best threshold)	**0.0%**
Most-common-value at our threshold	**0.0%**
Never answer	**0.0%**

cuad: 0.0% saved, precision n/a extractbench: 0.0% saved, precision n/a

Business case: how it moves if mistakes cost more



What it means

Automated with no human: 0.0% of fields; 100.0% go to a person

Tokens per document: 0 | USD per document: 0.0 | 0.26 s per document

Never automated: nested objects (all-or-nothing match) and any field whose confident answers fail the audit

Rollout: shadow mode (compare to humans) → auto-book the safest fields → widen as the calibration table holds

Honest limits and next steps

Limits

- Only ~42% of all fields have a value, so even a perfect system saves at most ~42% of the manual cost.

- Gold labels are noisy (e.g. CUAD 'Effective Date' sometimes labels the whole term sentence); F1>=0.8 is strict.

- A calibrator is only valid for the models it was fitted on: re-fit after changing the extractor or verifier (1 command).

- Scans without a vision-capable model fall back to human review: safe, but no saving.

Next steps

- Fine-tune the verifier on our own booked/rejected fields (the decision model is cheap to adapt).

- Learn per-field thresholds once enough shadow-mode data exists.

- Add layout-aware OCR zoom for low-quality scans and checkbox crops.

- Wire the review queue: show evidence + proof notes next to the model's value to make each review faster.